

Math 242, S16
Quiz 9

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Find the general solution of the given nonhomogeneous differential equation:

$$y'' - y' - 2y = 2x^2 + 3.$$

Solution: Step 1: Find the complementary solution $y_c(x)$

Characteristic equation:

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

$$r_1 = 2, r_2 = -1$$

$$\Rightarrow y_c(x) = C_1 e^{2x} + C_2 e^{-x}$$

Step 2: Find a particular solution $y_p(x)$

$$f(x) = 2x^2 + 3, f'(x) = 4x, f''(x) = 4$$

Thus we guess

$$y_p(x) = Ax^2 + Bx + C$$

$$y_p'(x) = 2Ax + B, y_p''(x) = 2A$$

Thus

$$\begin{aligned} y_p'' - y_p' - 2y_p &= (2A) - (2Ax + B) - 2(Ax^2 + Bx + C) \\ &= (-2A)x^2 + (-2A - 2B)x + (2A - B - 2C) \\ &= 2x^2 + 3 \end{aligned}$$

$$\Rightarrow \begin{cases} -2A = 2 \\ -2A - 2B = 0 \\ 2A - B - 2C = 3 \end{cases} \Rightarrow \begin{matrix} A = -1 \\ B = 1 \\ C = -3 \end{matrix} \Rightarrow y_p(x) = -x^2 + x - 3$$

Thus the general solution is given by

$$y(x) = y_c(x) + y_p(x) = C_1 e^{2x} + C_2 e^{-x} - x^2 + x - 3$$